

## MODULE 1 OF 6

# Foundations: Resources, Reserves and the Reporting Codes

*Resource Estimation & Geostatistics — a technical training course built on the live ANRML26-137 / EL-046017 datasets*

Before a single variogram is fitted, you must know exactly what you are trying to produce, what word you are allowed to put on it, and who is legally accountable for it. This module installs that framework. Everything in Modules 2 to 6 is machinery in service of the definitions you learn here.

## LEARNING OBJECTIVES

- State precisely what a Mineral Resource and an Ore Reserve are, and the test that separates them.
- Explain the CRIRSCO family of codes and the legal role of the Competent or Qualified Person.
- Place an estimate into the Inferred–Indicated–Measured framework and justify the category against evidence.
- Enumerate the modifying factors and explain how they convert a Resource into a Reserve.
- Define an Exploration Target and explain why Toro currently is one.

## 1 The question resource estimation answers

A mineral deposit is, commercially, a single quantity of money buried in a single volume of rock. The whole of resource estimation exists to answer one question precisely enough that capital will move on the answer.

### The central question

For a defined volume of ground: **how many tonnes are there, at what average grade, and with what level of confidence?** The third clause is not optional. An estimate without a stated confidence is not a resource estimate — it is a guess with decimal places.

Tonnes and grade are the two estimated quantities; confidence is the meta-statement about how far they can be trusted. Geostatistics (Modules 4–5) is the mathematics of that third clause. The two estimated quantities combine into the number that matters, contained metal:

$$M = T \times \bar{g} \quad (1.1)$$

Grade carries units that must be handled with discipline. The Toro laboratory reports tin as the oxide  $\text{SnO}_2$  in weight per cent; metal accounting and contracts use the element Sn. The conversion is a fixed stoichiometric factor:

$$\text{Sn} = \text{SnO}_2 \times \frac{M_{\text{Sn}}}{M_{\text{SnO}_2}} = \text{SnO}_2 \times \frac{118.71}{150.71} = 0.7877 \text{ SnO}_2 \quad (1.2)$$

A reported 6.04% SnO<sub>2</sub> in Toro pit DBP-002 is 4.76% Sn. Mixing the two is one of the most common and most expensive reporting errors in tin projects.

## 1.1 The exploration value chain

A resource estimate is one station on a chain running from a geological idea to a financed mine. Each transition is a *decision gate*: if results do not justify the next, larger tranche of capital, the project is downgraded, deferred or relinquished.

| Stage                    | Output                           | Toro status (May 2026)      |
|--------------------------|----------------------------------|-----------------------------|
| Desktop study            | Prospectivity model, targets     | Substantially complete      |
| Reconnaissance           | Field-confirmed anomalies        | Complete: 47 XRD, 40 XRF    |
| Systematic pitting       | Stratigraphy, Exploration Target | In progress                 |
| Resource drilling        | Drillhole database               | 13 priority collars defined |
| <b>Resource estimate</b> | Classified Resource statement    | Destination of this course  |
| Studies / Reserve        | Feasibility, an Ore Reserve      | Future                      |

## 2 Why there are codes at all

Public reporting of mineral assets is regulated because the history of not regulating it is a history of fraud. The **Poseidon bubble** (Australia, 1969–70) drove a nickel stock from under a dollar to GBP 280 and back on unregulated reporting. The **Bre-X fraud** (Busang, 1995–97) salted drill core with placer gold, claimed a fictitious 70-million-ounce deposit, and erased roughly USD 6 billion of value.

The Committee for Mineral Reserves International Reporting Standards maintains an international *template*. National codes implement it: **JORC** (Australasia), **NI 43-101** with the CIM Definition Standards (Canada), **SAMREC** (South Africa), **PERC** (Europe). They share one architecture, so a Resource is portable across exchanges. Nigeria has no domestic code, so a Toro estimate intended to raise capital will be reported under JORC or NI 43-101.

Three principles run through every code: **transparency** (the reader is given enough information not to be misled), **materiality** (all information a reasonable investor would require), and **competence** (a named, accountable signatory).

### The Competent / Qualified Person

A **Competent Person** (JORC) or **Qualified Person** (NI 43-101) takes personal, legal responsibility for the estimate. They must hold membership of a Recognised Professional Organisation and have at least five years of experience *relevant to the style of mineralisation under consideration*. That phrase matters for Toro: hard-rock gold experience does not qualify a person to sign an alluvial tin placer. COMEG registration and NMGS membership are the route to this status in the Nigerian context.

### 3 Mineral Resource versus Ore Reserve

A concentration of material of economic interest in or on the Earth's crust in such form, grade and quantity that there are *reasonable prospects for eventual economic extraction*. Defined on the basis of geological evidence and knowledge.

The economically mineable part of a Measured or Indicated Mineral Resource, demonstrated by at least a Pre-Feasibility Study that has applied the modifying factors.

Three consequences follow. A **Resource is geological; a Reserve is engineering plus economics. Resources are not a subset of Reserves — it is the reverse:** a Reserve is carved from the Measured and Indicated parts of a Resource, and an Inferred Resource can never become a Reserve without further drilling. And “**reasonable prospects for eventual economic extraction**” (**RPEEE**) **is a real, enforceable test:** you demonstrate it by constraining the estimate inside a conceptual mining envelope at a reasonable price and cut-off. Material that cannot sit within a plausible mining shape is not a Resource.

### 4 The classification framework

Within a Mineral Resource, confidence is graded into three categories — each a statement about how much geology has been *demonstrated* rather than *assumed*.

|                  |                                                                                                                                                   |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inferred</b>  | Estimated on limited evidence; geological and grade continuity <i>implied but not verified</i> . Cannot be converted to a Reserve.                |
| <b>Indicated</b> | Estimated with enough confidence to <i>reasonably assume</i> continuity. Supports a Pre-Feasibility Study; converts to a <b>Probable</b> Reserve. |
| <b>Measured</b>  | Estimated with enough confidence to <i>confirm</i> continuity. Supports a Feasibility Study; converts to a <b>Proven</b> Reserve.                 |

The codes express classification as a two-dimensional structure: one axis is geological confidence (raised by drilling and sampling), the other is the modifying factors (applied by study engineers). An Inferred Resource has no path straight to a Reserve — it must first be upgraded.

It is tempting to think the categories are defined by drill spacing alone (“50 m equals Measured”). They are not. Drill spacing is evidence, but the category is a *judgement about continuity* made by the Competent Person, supported by variography (Module 4), geological understanding and QAQC. Two deposits at identical spacing can warrant different categories.

### 5 The modifying factors

The modifying factors convert a geological Resource into an economically mineable Reserve. A code-compliant study addresses each one explicitly: **mining** (can it be mined, at what dilution and

recovery), **processing** (can the metal be liberated — cassiterite is gravity-recoverable, columbite needs magnetic separation), **economic** (does it pay at a realistic long-term tin price), **marketing** (concentrate specification and penalties), **infrastructure**, **legal/tenure**, **environmental**, and **social/governmental**. For the Toro alluvial target one factor demands early attention: the wash gravels are already mined by artisanal operators — simultaneously a confirmation of grade continuity and a serious social and tenure factor.

## 6 The Exploration Target — and why Toro is one

A statement of the *exploration potential* of a deposit, expressed as **ranges** of tonnage and grade. It must be clearly labelled as conceptual, state that there has been insufficient exploration to estimate a Mineral Resource, and not be capable of being misread as one.

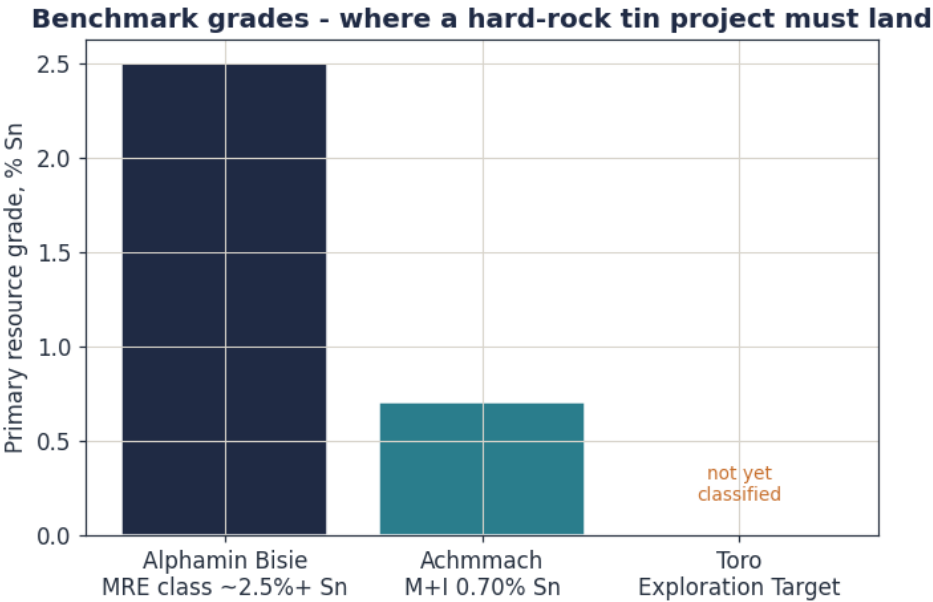
The current Toro grade data is a reconnaissance dataset: 40 XRF analyses collapsing to roughly **26 distinct pit locations** once laboratory mineral-separation fractions are removed (Module 2). Those pits spread across three drainages over several kilometres. Two facts follow. First, the sampling density cannot support an assumption of grade continuity at any mining scale — so no category above Exploration Target is defensible. Second, the panned-concentrate sampling medium does not by itself give an in-situ g/m<sup>3</sup> grade. Toro is an Exploration Target because the data type and density make it one, and only drilling and systematic pitting can change that.

## 7 The Achmmach benchmark

Atlantic Tin's Achmmach project in Morocco reached, after roughly 18 years, a Measured and Indicated Resource of **22.4 Mt at 0.70% Sn** (about 156,000 t contained tin) and Proven and Probable Reserves of **7.0 Mt at 0.82% Sn** (about 58,000 t contained tin). The Resource is the larger, geologically defined quantity; the Reserve is the smaller part a feasibility study showed could be mined, at slightly higher grade because marginal material fell below cut-off. That is the Resource-to-Reserve relationship in real numbers. Toro's task over the next five to seven years is to earn the right to write a comparable pair of lines.

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*Module 1 of the Toro Resource Estimation & Geostatistics course. Next: Module 2, Sampling, Data and Database Integrity.*



**Figure 1** – Where a hard-rock tin project must land. Toro’s primary target has no classified grade yet — it is an Exploration Target.